



Business Requirement

Version *[1.0.0]*

New RM Selection

Modul: Master Data

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1. PENDAHULUAN

1.1 Tujuan Dokumen

Dokumen ini menjelaskan spesifikasi fungsional dari Modul Master Data pada sistem RM Selection (Raw Material Selection) dengan struktur database aktual. Master data ini dibutuhkan untuk modul transaksional seperti New RM Sample.

1.2 Ruang Lingkup

Sistem ini mencakup 6 form/modul master utama: - BTP (Bahan Tambahan Pangan) - BTP Function - RM Category - RM Sub Group - Master UOM (Unit of Measure) - Test Parameter Conversion

1.3 Stakeholder

- R&D Department: Pengguna utama untuk setup kategori dan parameter uji.
- QA/QC Department: Review dan referensi standar parameter pengujian.
- Purchasing Department: Referensi RM yang terdaftar.
- Admin/IT: Maintenance system.

2. RINGKASAN BUSINESS FLOW

2.1 Proses As-Is (Legacy System)

Pengelolaan spesifikasi RM, BTP, dan Unit menggunakan tabel-tabel terpisah yang belum terpusat sepenuhnya, sehingga maintenance data (Create, Update, Delete) membutuhkan effort administrasi database secara manual, tanpa antarmuka yang tersentralisasi.

2.2 Proses To-Be (New System)

Semua master data dikelola melalui antarmuka web yang terpusat dengan single entry point. Menggunakan relasi antartabel dengan Junction tables (contoh: M_BTP_DETAIL) untuk menjaga integritas relasi Many-to-Many tanpa redundansi. Modul ini menyediakan data lookup (LOV) bagi seluruh aplikasi RM Selection.

3. SPESIFIKASI FUNGSIONAL

3.1 BTP (Bahan Tambahan Pangan)

3.1.1 Deskripsi

Navigation: Sidebar > Master Data > BTP Halaman ini menampilkan daftar BTP (Bahan Tambahan Pangan) dalam bentuk tabel. Memungkinkan admin untuk menambahkan (Create), mengubah (Edit), dan menghapus secara soft-delete (bitActive=0) data BTP.

3.1.2 Screenshot

IDC System

Dashboard

APPS

RM Selection

Master

BTP

BTP Function

RM Sub Group

RM Category

Master UOM

Parameter

Test Parameter Conversion

Transaction

Packforces

Search...

Administrator

Master / BTP

Create New

Data List

Show 10 entries

Search

BTP CODE	BTP NAME	ACTIVE	ACTION
INS_1000	Kurkumin Cl. No. 75300 (Curcumin)	☒	<a>Edit <a>Function
INS_101	Riboflavin (Riboflavins)	☒	<a>Edit <a>Function
INS_1010	Riboflavin (sintetik) (Riboflavin, synthetic)	☒	<a>Edit <a>Function
INS_1010a	Riboflavin 5 natrium fosfat (Riboflavin 5 phosphate sodium)	☒	<a>Edit <a>Function
INS_1010a	Riboflavin (Bacillus subtilis) [Riboflavin (Bacillus subtilis)]	☒	<a>Edit <a>Function
INS_102	Tartrazin Cl. No. 19140 (Tartrazine)	☒	<a>Edit <a>Function
INS_104	Kuning kuinolin Cl. No. 47005 (Quinoline yellow)	☒	<a>Edit <a>Function
INS_110	Kuning FCF Cl. No. 15985 (Sunset yellow FCF)	☒	<a>Edit <a>Function
INS_1100	aAmilase (karbhidrase) dari Bacillus licheniformis (alphaAmylase from Bacillus licheniformis (carbohydrase)	☒	<a>Edit <a>Function
INS_1100	aAmilase dari Aspergillus oryzae, var (alphaAmylase from Aspergillus oryzae, var.)	☒	<a>Edit <a>Function

Showing 1 to 10 of 242 entries

Previous

1

2

3

4

5

...

25

Next

IDC System

Dashboard

APPS

RM Selection

Packforces

Search...

Administrator

Master / BTP / Detail

BTP Information

BTP Code

BTP Name

Active ☒

Save Back

3.1.3 Functional Description

Halaman Detail / Create:

No	Element	Type	Action / Behavior	Data Source
1	BTP Code	Text Input	Kode unik BTP, alfanumerik max 50 char	M_BTP.txtBtpCode
2	BTP Name	Text Input	Nama BTP, min 3 max 200 char	M_BTP.txtBtpName

No	Element	Type	Action / Behavior	Data Source
3	Active	Checkbox	Status aktif BTP (default checked/true)	M_BTP.bitActive
4	Save	Button	Simpan perubahan ke DB dengan validasi unik	-
5	Cancel	Button	Batal kembali ke Index	-

3.1.4 Business Rules & Validation

- **RBAC & Status Permission:** Hanya user dengan role Administrator yang dapat Edit/Delete BTP master data.
- **Validasi Field:** BTP Code (Required, Unique, tanpa spasi), BTP Name (Required).
- **Business Logic:** BTP yang sudah digunakan di M_BTP_DETAIL tidak dapat di-soft delete atau diubah kodenya.

3.2 BTP Function

3.2.1 Deskripsi

Navigation: Sidebar > Master Data > BTP Function Halaman untuk mendaftarkan fungsi dari masing-masing BTP, mengacu ke standard BPOM (PERKA BPOM).

3.2.2 Screenshot

The screenshot displays the IDC System interface. On the left is a sidebar with navigation options: Dashboard, APPS, RM Selection, Master, BTP, BTP Function (selected), RM Sub Group, RM Category, Master UOM, Parameter, Test Parameter Conversion, Transaction, and Packforces. The main content area is titled 'Master / BTP Function' and includes a '+ Create New' button. Below this is a 'Data List' table with columns: FUNCTION CODE, FUNCTION NAME, ACTIVE, and ACTION. The table contains 10 entries, each representing a BPOM regulation (PERKA BPOM) and its corresponding function name. The 'ACTIVE' column shows green checkmarks for all entries. The 'ACTION' column contains edit and delete icons. At the bottom of the table, it indicates 'Showing 1 to 10 of 26 entries' and includes pagination controls (Previous, 1, 2, 3, Next).

FUNCTION CODE	FUNCTION NAME	ACTIVE	ACTION
PERKA BPOM 07-2013	PERLAKUAN TEPUNG	✓	[Edit] [Delete]
PERKA BPOM 09-2013	PENGERAS	✓	[Edit] [Delete]
PERKA BPOM 10-2013	ANTIKEPAL	✓	[Edit] [Delete]
PERKA BPOM 11-2013	PENGEMBANG	✓	[Edit] [Delete]
PERKA BPOM 12-2013	PELAPIS	✓	[Edit] [Delete]
PERKA BPOM 13-2013	ANTIBUJI	✓	[Edit] [Delete]
PERKA BPOM 14-2013	PROPELAN	✓	[Edit] [Delete]
PERKA BPOM 15-2013	PENGENTAL	✓	[Edit] [Delete]
PERKA BPOM 16-2013	GARAM PENGEMULSI	✓	[Edit] [Delete]
PERKA BPOM 17-2013	GAS UNTUK KEMASAN	✓	[Edit] [Delete]

3.2.3 Functional Description

Halaman Detail:

No	Element	Type	Action / Behavior	Data Source
1	Function Code	Text Input	Kode fungsi unik mengacu PERKA BPOM	M_BTP_FUNCTION.txtFunctionCode
2	Function Name	Text Input	Nama lengkap fungsi	M_BTP_FUNCTION.txtFunctionName
3	Active	Checkbox	Status fungsi (aktif/nonaktif)	M_BTP_FUNCTION.bitActive

3.2.4 Business Rules & Validation

- **Validasi Field:** Code mengikuti format BPOM (contoh: PERKA BPOM 07-2013).
- **Business Logic:** Fungsi BTP yang sudah di-mapping oleh BTP tertentu tidak dapat dihapus.

3.3 RM Category

3.3.1 Deskripsi

Navigation: Sidebar > Master Data > RM Category Halaman untuk manajemen kategori utama bahan baku.

3.3.2 Screenshot

IDC System

Dashboard

APPS

- RM Selection
- Master
- BTP
- BTP Function
- RM Sub Group
- RM Category**
- Master UOM
- Parameter
- Test Parameter Conversion
- Transaction
- Packforces

Search...

Administrator

Master / RM Category

Create New

Data List

Show 10 entries

Search

RM CATEGORY CODE	RM CATEGORY NAME	ACTIVE	ACTION
Filter...	Filter...	Filter...	
A0	Dairy and its derivatives	☒	Edit Sub Group
A0	Dairy Products	☒	Edit Sub Group
A1	Dairy Products	☒	Edit Sub Group
B0	Carbohydrate (non dairy) and its Derivatives	☒	Edit Sub Group
B0	Carbohydrate, sugar, & Fiber	☒	Edit Sub Group
B1	Carbohydrate (non dairy) and its Derivatives	☒	Edit Sub Group
B1	Carbohydrate, sugar, & Fiber	☒	Edit Sub Group
C0	Fat (non dairy) and its derivatives	☒	Edit Sub Group
C1	Fat	☒	Edit Sub Group
D0	Protein (non dairy) and its derivatives	☒	Edit Sub Group

Showing 1 to 10 of 31 entries

Previous 1 2 3 4 Next

IDC System

Dashboard

APPS

- RM Selection
- Packforces

Search...

Administrator

Master / RM Category / Detail

RM Category Information

RM Category Code:

RM Category Name:

Critical: ☐

Apply Product: ☐

Active: ☒

[Save](#) [Back](#)

3.3.3 Functional Description

No	Element	Type	Action / Behavior	Data Source
1	Category Code	Text Input	Kode kategori unik	M_RM_CATEGORY.txtRMCategoryCode
2	Category Name	Text Input	Nama kategori bahan	M_RM_CATEGORY.txtRMCategoryName
3	Critical Flag	Checkbox	Menandakan kategori raw material critical	M_RM_CATEGORY.bitCritical
4	Apply Product	Checkbox	Flag apakah diaplikasikan ke product	M_RM_CATEGORY.bitApplyProduct
5	Active	Checkbox	Status kategori	M_RM_CATEGORY.bitActive

3.3.4 Business Rules & Validation

- Kategori berhubungan dengan sub group melalui M_RM_CATEGORY_DETAIL.

3.4 RM Sub Group

3.4.1 Deskripsi

Navigation: Sidebar > Master Data > RM Sub Group Halaman mengatur pengelompokan sub-tipe bahan untuk RM Category. Sub group tidak menempel langsung ke RM_CATEGORY tetapi via Junction tabel agar satu sub-group bisa dipakai multi-category.

3.4.2 Screenshot

The screenshot displays the IDC System interface for the 'Master / RM Sub Group' section. The left sidebar shows the navigation menu with 'RM Sub Group' highlighted. The main content area features a 'Data List' table with columns for 'RM SUB GROUP CODE', 'RM SUB GROUP NAME', 'ACTIVE', and 'ACTION'. The table lists 10 entries, each with a unique code and name, and an 'ACTIVE' status indicated by a green checkmark. The 'ACTION' column contains a blue pencil icon for editing. The table is paginated, showing 'Showing 1 to 10 of 232 entries' and a 'Previous' button.

RM SUB GROUP CODE	RM SUB GROUP NAME	ACTIVE	ACTION
AAC	Amino Acid	✓	✎
AAC	Amino Acid	✓	✎
AAD	ARA and DHA	✓	✎
AAD	ARA and DHA	✓	✎
ACA	Anti Caking	✓	✎
ACA	Anti Caking	✓	✎
ACO	Anti Coagulant	✓	✎
ACO	Anti Coagulant	✓	✎
ALA	Alpha Lactalbumin	✓	✎
ALA	Alpha Lactalbumin	✓	✎



RM Sub Group Information

RM Sub Group Code

RM Sub Group Code

RM Sub Group Name

RM Sub Group Name

Active



Save

Back

3.4.3 Functional Description

No	Element	Type	Action / Behavior	Data Source
1	Sub Group Code	Text Input	Kode unik	M_RM_SUB_GROUP.txtRMSubGroupCode
2	Sub Group Name	Text Input	Nama dari sub group bahan	M_RM_SUB_GROUP.txtRMSubGroupName
3	Active	Checkbox	Status	M_RM_SUB_GROUP.bitActive

3.4.4 Business Rules & Validation

- **Mapping Constraint:** Dikelola secara terpisah dari RM Category.

3.5 Master UOM (Unit of Measure)

3.5.1 Deskripsi

Navigation: Sidebar > Master Data > UOM Halaman manajemen satuan ukuran (Kg, Litre, Gram, dll).

3.5.2 Screenshot

IDC System

Dashboard

APPS

RM Selection

Master

BTP

BTP Function

RM Sub Group

RM Category

Master UOM

Parameter

Test Parameter Conversion

Transaction

Packforces

Search...

Administrator

Master / Master UOM

Create New

Data List

Show 10 entries

Search

UOM CODE	UOM NAME	ACTIVE	ACTION
Filter...	Filter...	Filter...	
%	Prosentase	☒	Edit
% b/b	% b/b	☒	Edit
% b/v	% b/v	☒	Edit
%wv	%wv	☒	Edit
/0.01 g	/0.01 g	☒	Edit
/0.1g	/0.1g	☒	Edit
/10 gr	/10 gr	☒	Edit
/100 gr	/100 gr	☒	Edit
/25 gr	/25 gr	☒	Edit

Showing 1 to 10 of 125 entries

Previous

1

2

3

4

5

13

Next

IDC System

Dashboard

APPS

RM Selection

Packforces

Search...

Administrator

Master / Master UOM / Detail

UOM Information

UOM Code

UOM Code

UOM Name

UOM Name

Active

☒

Save

Back

3.5.3 Functional Description

No	Element	Type	Action / Behavior	Data Source
1	UOM Code	Text Input	Kode unit satuan	M_UNIT.txtUnitCode
2	UOM Name	Text Input	Deskripsi satuan	M_UNIT.txtUnitName
3	Active	Checkbox	Status satuan	M_UNIT.bitActive

3.6 Test Parameter Conversion

3.6.1 Deskripsi

Navigation: Sidebar > Master Data > Test Parameter Conversion Halaman untuk setup rumus konversi evaluasi tes RM dari nilai raw lab menjadi final standar (mis. Protein %).

3.6.2 Screenshot

The screenshot shows the IDC System interface. On the left is a sidebar with navigation options: Dashboard, APPS, RM Selection, Master, BTP, BTP Function, RM Sub Group, RM Category, Master UOM, Parameter (highlighted), Test Parameter Conversion, Transaction, and Packforces. The main content area is titled 'Master / Parameter' and features a '+ Create New' button. Below this is a 'Data List' section with a search bar and a table of parameters. The table has columns for PARAMETER CODE, PARAMETER NAME, ACTIVE, and ACTION. It lists 10 parameters, including Moisture, Ash, Carbohydrate, Total Sugar, Sucrose, Lactose, Galactose, Dietary Fiber, Protein, and Phospholipid. At the bottom, it indicates 'Showing 1 to 10 of 200 entries' and includes pagination controls.

PARAMETER CODE	PARAMETER NAME	ACTIVE	ACTION
1	Moisture	☒	Edit
2	Ash	☒	Edit
3	Carbohydrate	☒	Edit
4	Total Sugar	☒	Edit
5	Sucrose	☒	Edit
6	Lactose	☒	Edit
7	Galactose	☒	Edit
8	Dietary Fiber	☒	Edit
9	Protein	☒	Edit
10	Phospholipid	☒	Edit

The screenshot shows the 'Master / Parameter / Detail' page in the IDC System. It contains a form for 'Parameter Information' with fields for Parameter Code (901), Parameter Name (P.aeruginosa), Parameter AKASIA, Unit of Measure (Select UOM...), Flag (Select Flag...), and Active status (checked). There are 'Save' and 'Back' buttons at the bottom of the form.

Parameter Information

Parameter Code: 901

Parameter Name: P.aeruginosa

Parameter AKASIA:

Unit of Measure: [Select UOM...](#)

Flag: [Select Flag...](#)

Active: ☒ Contaminant ☐

[Save](#) [Back](#)

3.6.3 Functional Description

No	Element	Type	Action / Behavior	Data Source
1	Factor Number	Decimal	Sebagai angka penyebut rumus konversi	M_TEST_PARAMETER_CONVERSION.decFactorNumber
2	txtFactor	Text Input	Pengali dari nilai test	M_TEST_PARAMETER_CONVERSION.txtFactor
3	Target	Data Grid	Target per grup RM	M_TARGET_PER_RM_CATEGORY

3.6.4 Business Rules & Validation

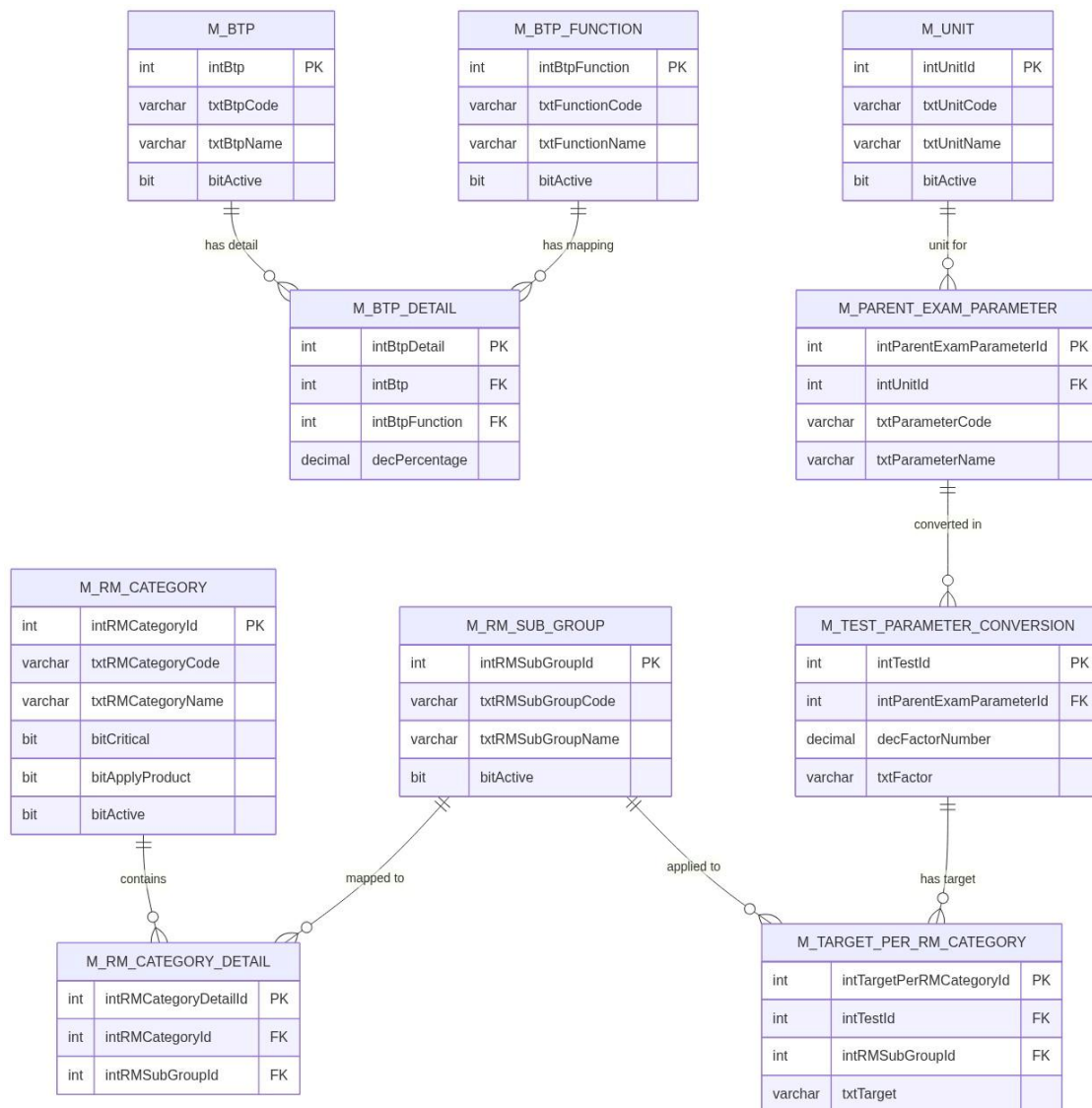
- **Formula Hitung:** Hasil Konversi = (Nilai Test × txtFactor) / decFactorNumber.

4. STRUKTUR DATABASE

4.1 Konvensi Penamaan

- Tabel Master menggunakan awalan M_. Relasi many-to-many dilayani oleh tabel junction dengan imbuhan _DETAIL atau nama penghubung.
- Penamaan field mengikuti standar prefix Hungary (int untuk nomor/ID autonumber, txt untuk teks varchar).

4.2 Entity Relationship Diagram (ERD)



Setiap sub-modul master memiliki dependency key ke tabel junction yang mengatur parameter-parameter khusus.

4.3 Tabel Detail Master & Junction

Nama Tabel	Tipe	Primary Key	Keterangan
M_BTP	Master	intBtp	Bahan Tambahan Pangan
M_BTP_FUNCTION	Master	intBtpFunction	Master Fungsi BTP
M_BTP_DETAIL	Junction	intBtpDetail	(intBtp, intBtpFunction) relation
M_RM_CATEGORY	Master	intRMCATEGORYId	Master kategori RM
M_RM_SUB_GROUP	Master	intRMSubGroupId	Master sub grup RM
M_RM_CATEGORY_DETAIL	Junction	intRMCATEGORYDetailId	Menghubungkan Kategori dengan Subgroup
M_UNIT	Master	intUnitId	Master satuan (UOM)
M_PARENT_EXAM_PARAMETER	Master	intParentExamParameterId	Master tipe uji (FK: intUnitId)
M_TEST_PARAMETER_CONVERSION	Trans	intTestId	Set nilai convert untuk parameter pengujian
M_TARGET_PER_RM_CATEGORY	Junction	intTargetPerRMCATEGORY	Target untuk tiap SubGroup pengujian